

Understanding Human Motor Control

In my postdoc with Prof. Marcia O'Malley, I investigated human motor control during dynamic manual tasks and how haptic feedback closes the internal sensory-motor control loop for skill acquisition, retention, transfer and adaptation.

Notable Publications:

- Israr, A., Li, Y., Patoglu, V., & O'Malley, M. K. (2008). Passive and active discrimination of natural frequency of virtual dynamic systems. IEEE transactions on haptics, 2(1), 40-51. [\[Link\]](#)

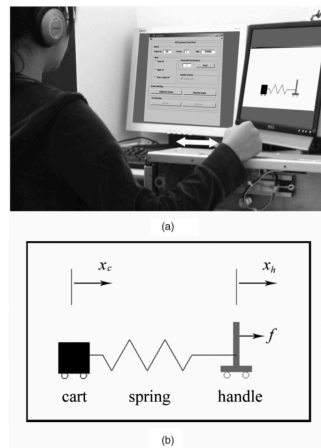


Fig. 2. Discrimination experiment setup, (a) showing the 1-DOF haptic device held by the participant and dual screen display, (b) shows the graphical display of the virtual environment consisting of a cart and handle system connected by a spring.

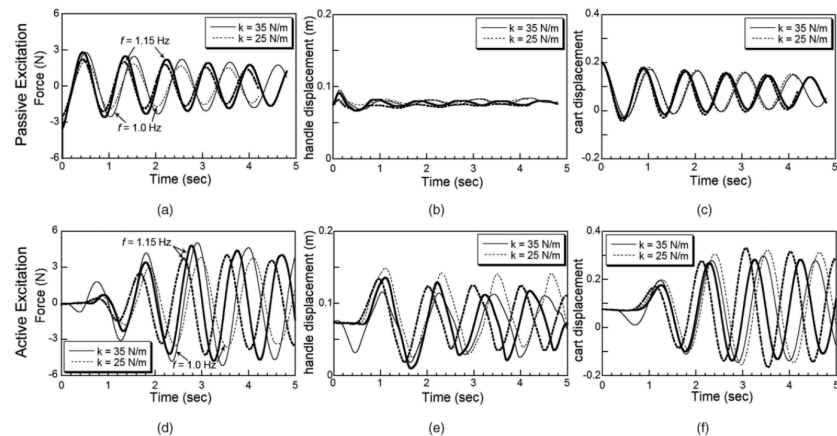


Fig. 6. Typical rendered force and handle and cart displacement profiles at 1 Hz reference natural frequency in (a), (b), (c) passive and (d), (e), (f) active modes.

- Israr, A., Kapson, H., Patoglu, V., & O'Malley, M. K. (2009, March). Effects of magnitude and phase cues on human motor adaptation. In World Haptics 2009-Third Joint EuroHaptics conference and Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (pp. 344-349). IEEE. [\[Link\]](#)

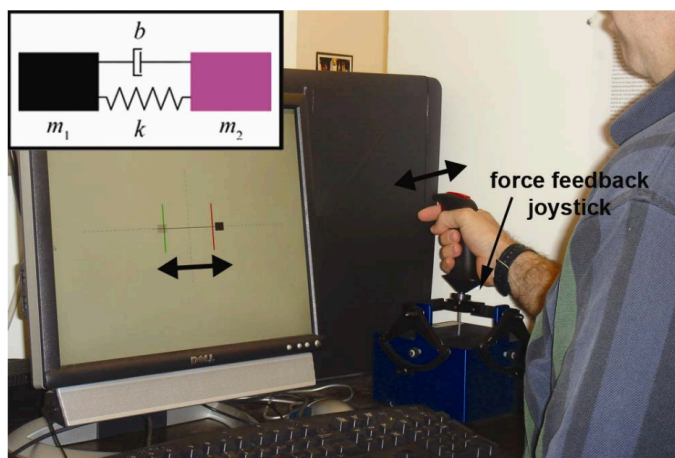


Figure 1. Experimental setup and virtual environment. (inset virtual two mass system connected by a spring-damper system.

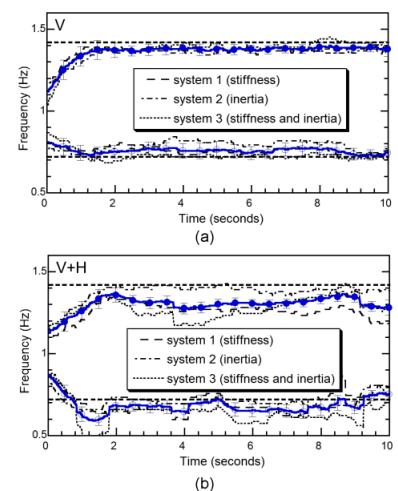


Figure 4. Time-frequency profiles of hand displacement in C2 (a) V feedback and (b) V+H feedback.

- Erdogan, A., Israr, A., O'Malley, M. K., & Patoglu, V. (2011, June). Rate of human motor adaptation under varying system dynamics. In 2011 IEEE World Haptics Conference (pp. 167-172). IEEE. [\[Link\]](#)
- Huegel, J. C., Celik, O., Israr, A., & O'Malley, M. K. (2009). Expertise-based performance measures in a virtual training environment. Presence: Teleoperators and Virtual Environments, 18(6), 449-467. [\[Link\]](#)

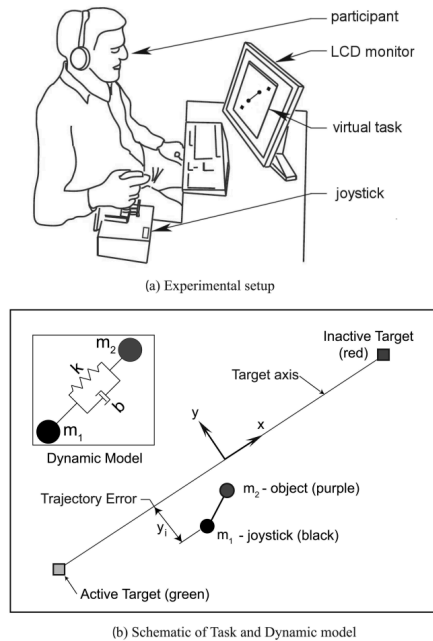


Figure 1. (a) Experimental setup includes a force feedback joystick and the graphical interface. (b) Schematic shows the target hitting task and its coordinate frame as well as the modeled 4-DOF second order dynamic system.

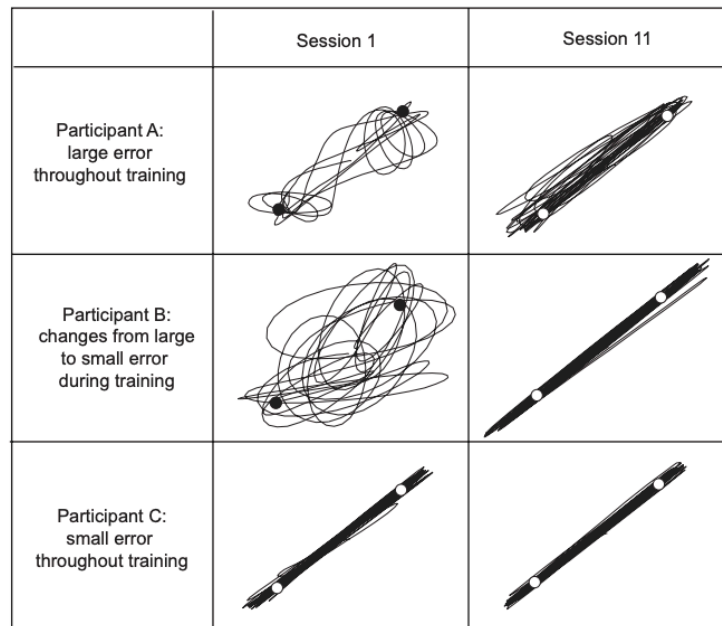


Figure 3. Sample traces for three typical participants shows varying improvements and performance differences.

- Israr, A., Eng, D. P., Vedam, S. S., & O'Malley, M. K. (2009, March). A low cost vibrotactile array to manage respiratory motion. In World Haptics 2009-Third Joint EuroHaptics conference and Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (pp. 619-620). IEEE. [\[Link\]](#)

We developed a tactile Respiratory Management System (tRMS) to maintain uniform and steady breathing patterns for lung cancer patients undergoing radiotherapy treatment using tactile feedback. A well known “gated-therapy” technique targets infected tumors during either the full inhale or the full exhale posture. In order to reduce the exposure time and increase the efficacy of treatment, patients need to maintain a normal and/or predefined chest motion during treatments. The system consists of an array of tactile transducers arranged on the forearm.

